A. Requirement of controller structure

Requirements

(i) Zero steady-state error for constant friction torque of rotary arm.

(ii) 2% settling time of the closed-loop system should be less than 0.5sec.

(iii) OS% should be no more than 20%

Analyze

For (ii):

$$t_s(\pm 2\%) = \frac{4}{\xi\omega_n} = \frac{4}{\sigma} \leqslant 0.5$$

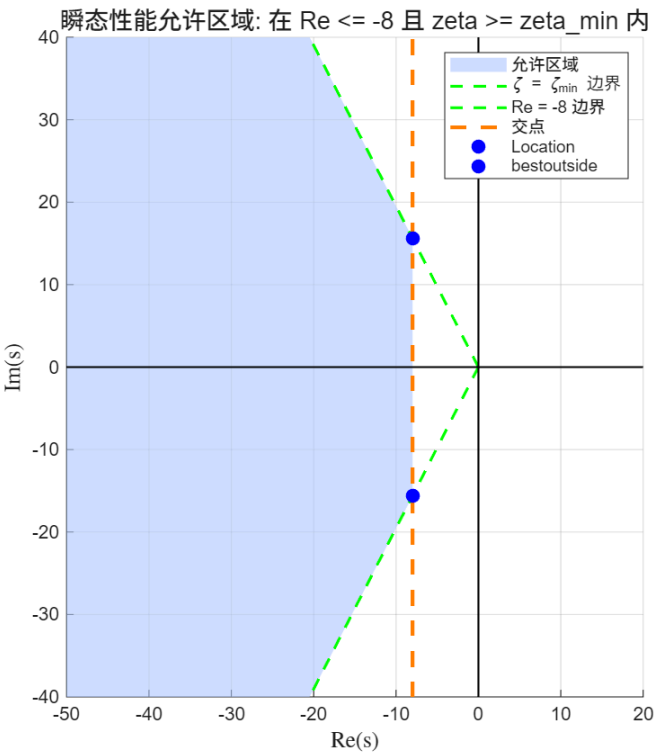
$$\therefore \sigma \geqslant 8$$

For (iii):

$$OS\% = 100\% \times e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \leqslant 20\%$$

$$\zeta \geqslant \zeta_{min} = \sqrt{\frac{1}{\frac{\pi}{ln(5)}^2+1}} \approx 0.4559459$$

$$\varphi \leqslant \arccos \zeta_{min} \approx 62.87^\circ$$



B. Proper PID

$$C(s) = K_p + \frac{K_i}{s} + \frac{K_d s}{\tau s + 1}$$

Placing poles at [-10+10j, -10-10j, -50, -50], so the expected polynomial is
$s^4 + 120s^3 + 4700s^2 + 70000s + 500000$

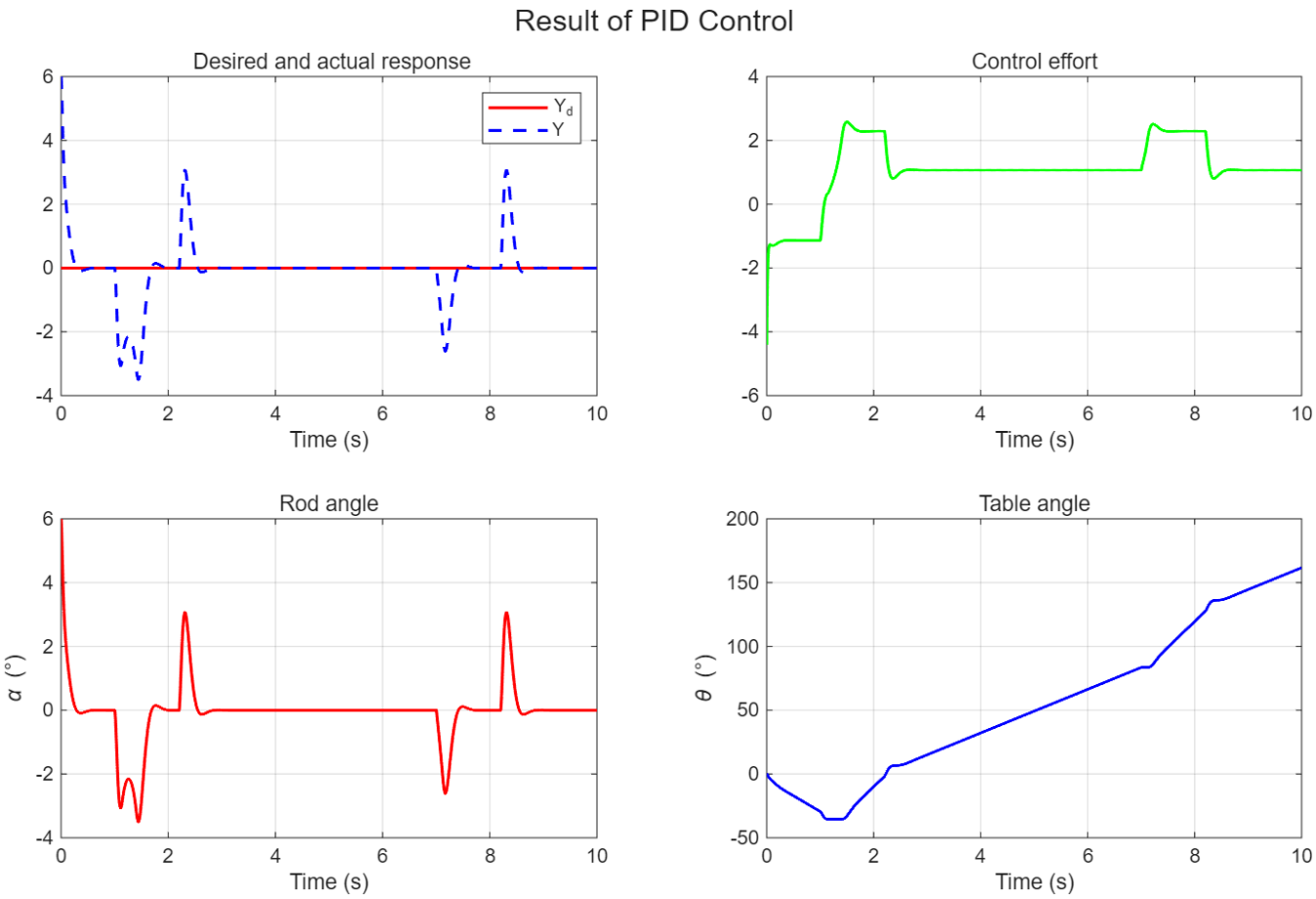
Design results

$\tau_D = 0.0111362, K_P = 16.564, K_I = 137.031, K_D = 0.284031$

$C(s) = \frac{0.4685s^2+18.09s+137}{0.01114s^2+s}$

$T(s) = \frac{2095s^2+8.091\times10^4 s+6.129\times10^5}{s^4+120s^3+4700s^2+7\times10^4 s+5\times10^5}$

Verification by Simulink



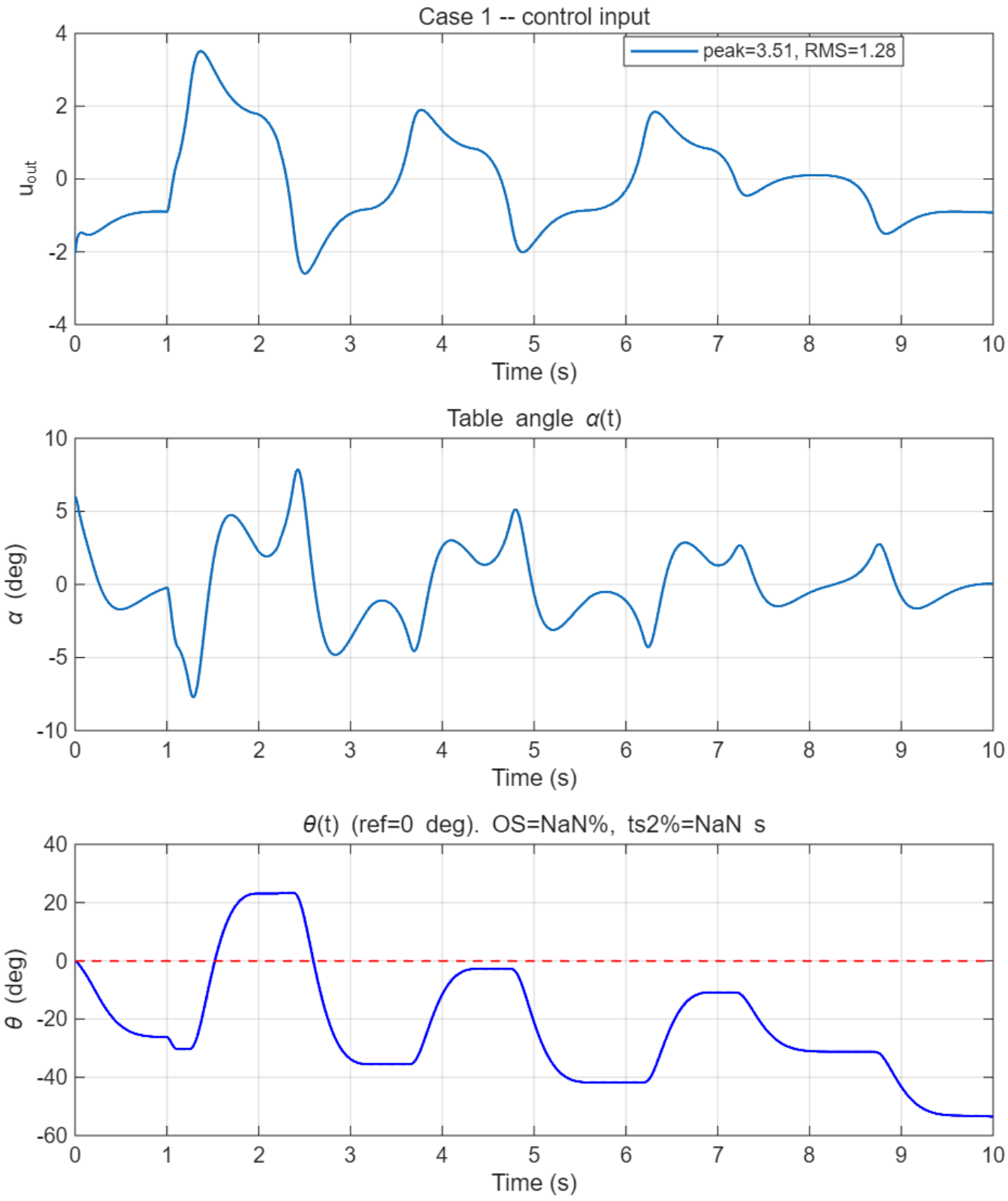
Performance requirements are satisfied.

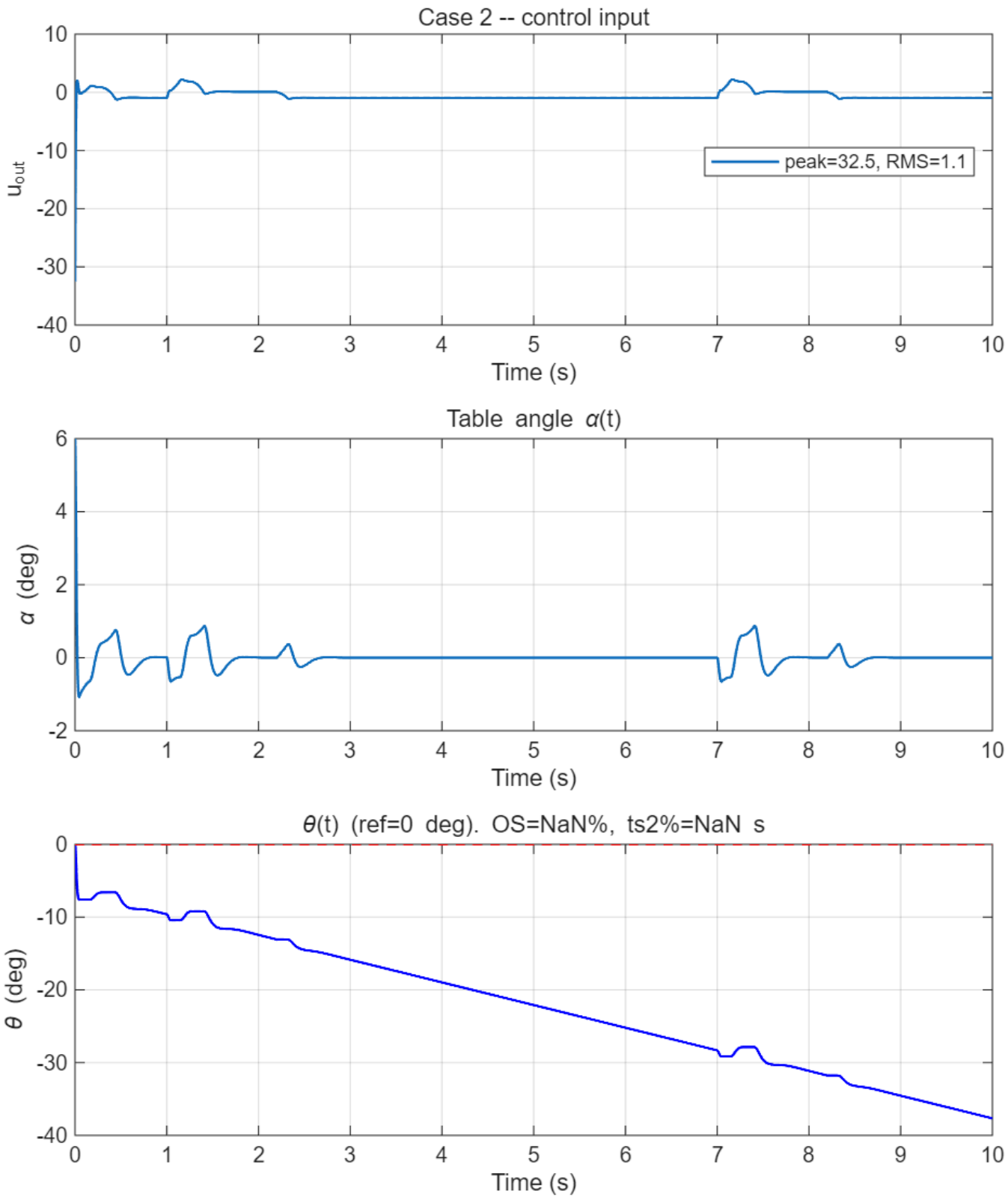
C. General Controller

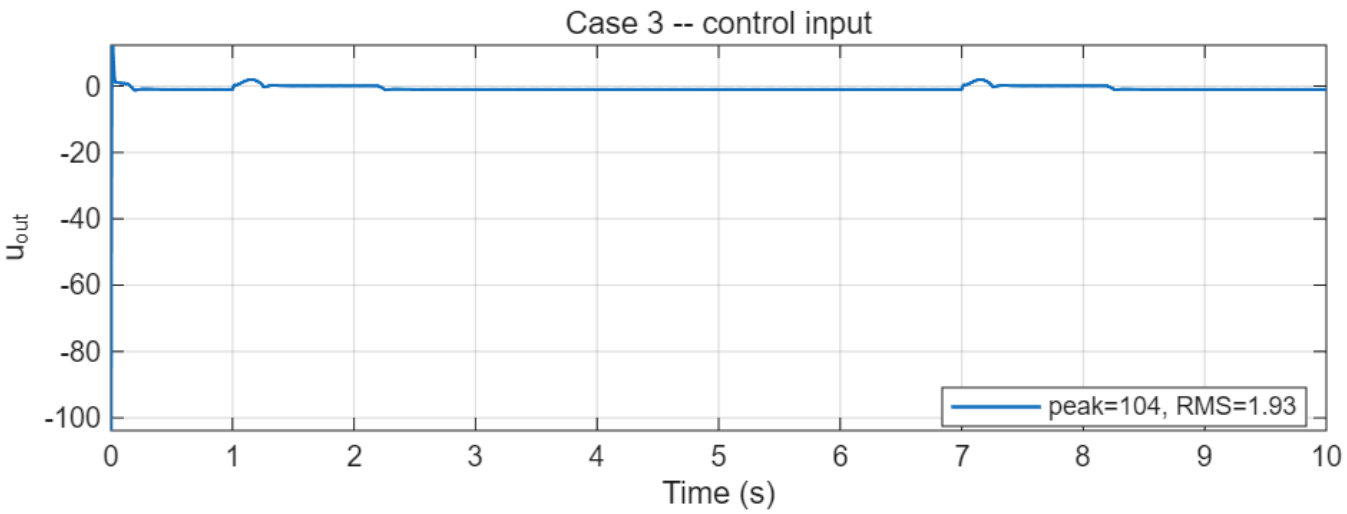
Cases for controller design with pole placement method

Case	Closed-loop bandwidth	Initial rod angle
Case 1	$\omega_{cl} = 5$, $\zeta_{cl} = 0.707$	$\alpha_0 = 6°$
Case 2	$\omega_{cl} = 15$, $\zeta_{cl} = 0.707$	$\alpha_0 = 6°$
Case 3	$\omega_{cl} = 25$, $\zeta_{cl} = 0.707$	$\alpha_0 = 6°$
Case 4	$\omega_{cl} = 5$, $\zeta_{cl} = 0.707$	$\alpha_0 = 10°$
Case 5	$\omega_{cl} = 15$, $\zeta_{cl} = 0.707$	$\alpha_0 = 10°$
Case 6	$\omega_{cl} = 25$, $\zeta_{cl} = 0.707$	$\alpha_0 = 10°$

Verification by Simulink




Case 2 -- control input
u_out
peak=32.5, RMS=1.1
Time (s)
Table angle α(t)
α (deg)
Time (s)
θ(t) (ref=0 deg). OS=NaN%, ts2%=NaN s
θ (deg)
Time (s)


Case 3 -- control input
u_out
0
-20
-40
-60
-80
-100
peak=104, RMS=1.93
0 1 2 3 4 5 6 7 8 9 10
Time (s)

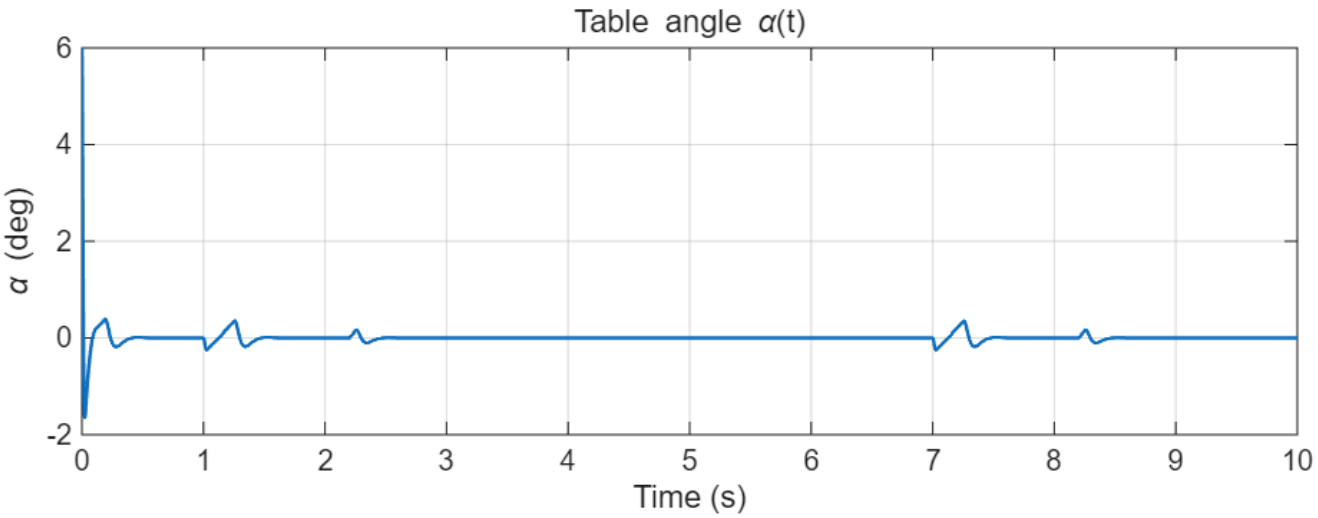
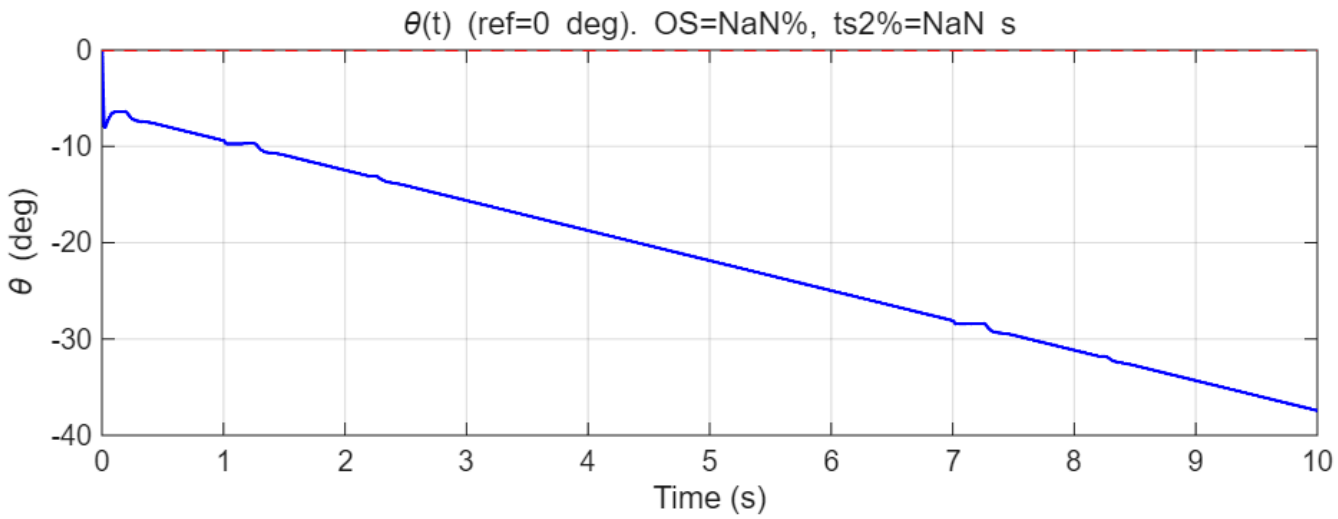
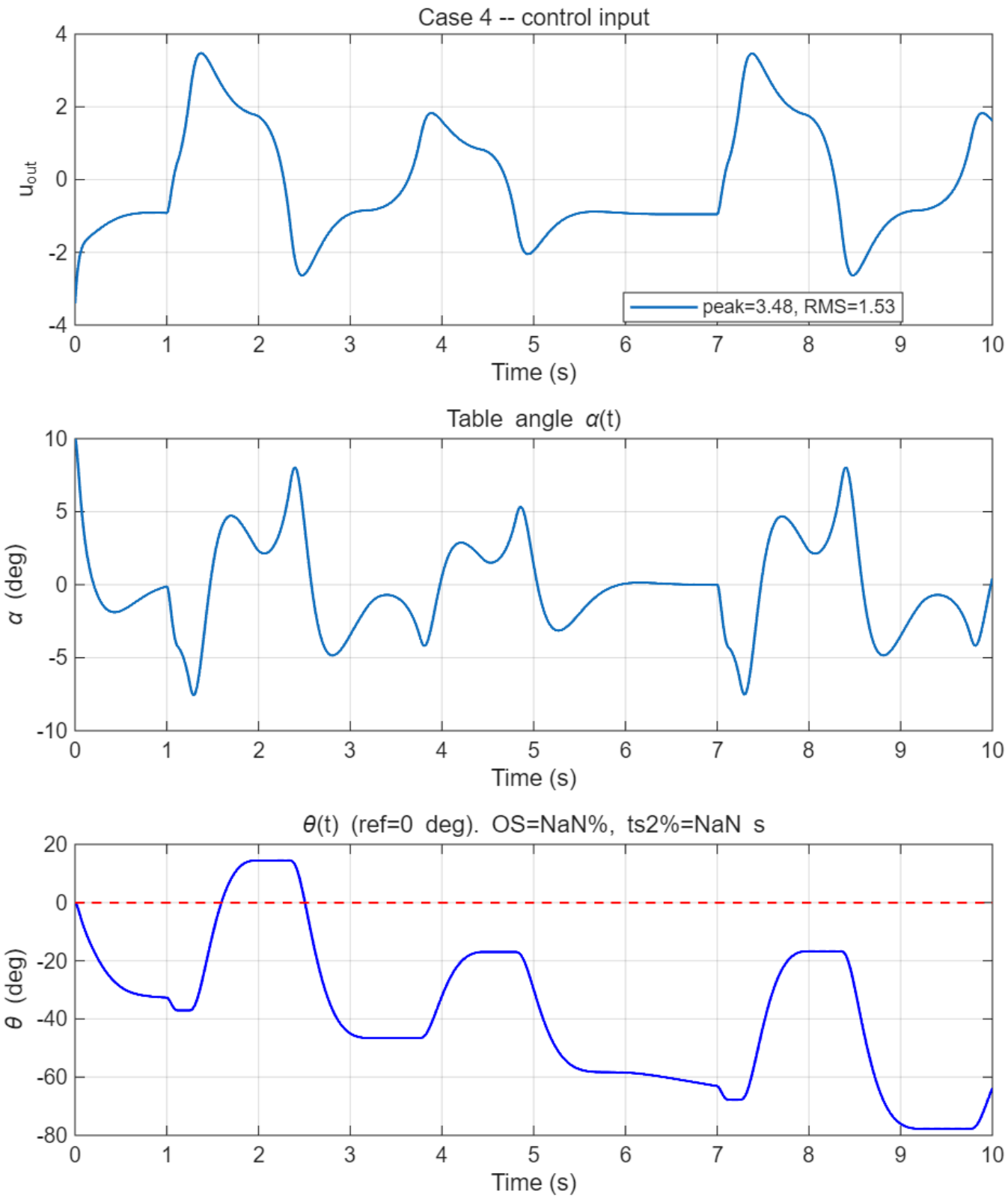
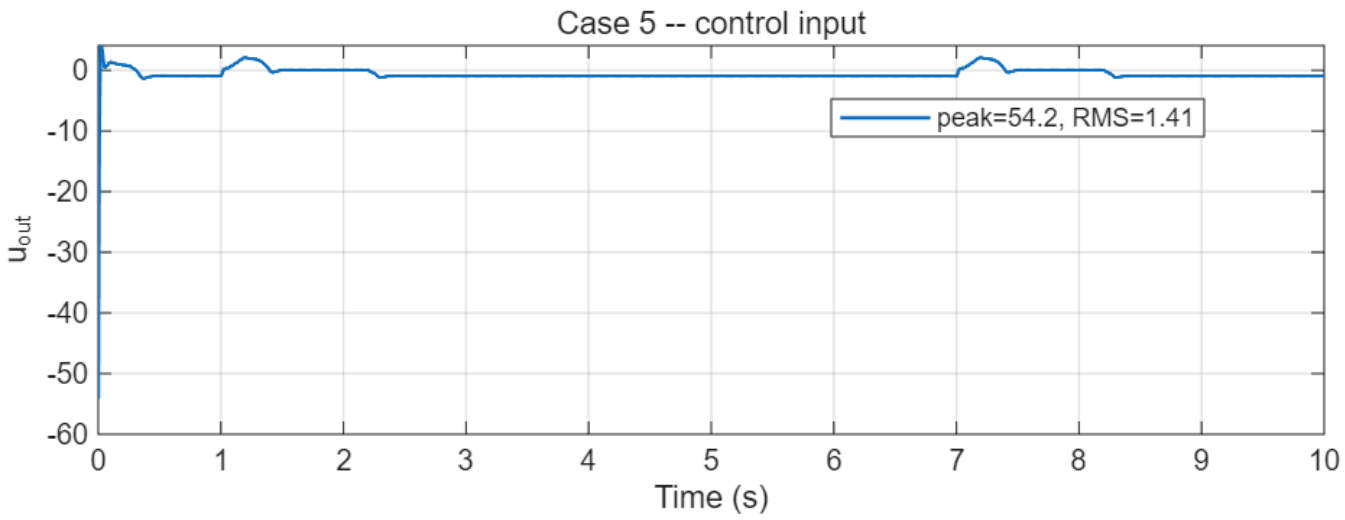


Table angle α(t)
α (deg)
6
4
2
0
-2
0 1 2 3 4 5 6 7 8 9 10
Time (s)


θ(t) (ref=0 deg). OS=NaN%, ts2%=NaN s
θ (deg)
0
-10
-20
-30
-40
0 1 2 3 4 5 6 7 8 9 10
Time (s)


Case 4 -- control input
u_out
peak=3.48, RMS=1.53
Time (s)
Table angle α(t)
α (deg)
Time (s)
θ(t) (ref=0 deg). OS=NaN%, ts2%=NaN s
θ (deg)
Time (s)


Case 5 -- control input
peak=54.2, RMS=1.41
u_out
0
-10
-20
-30
-40
-50
-60
0 1 2 3 4 5 6 7 8 9 10
Time (s)

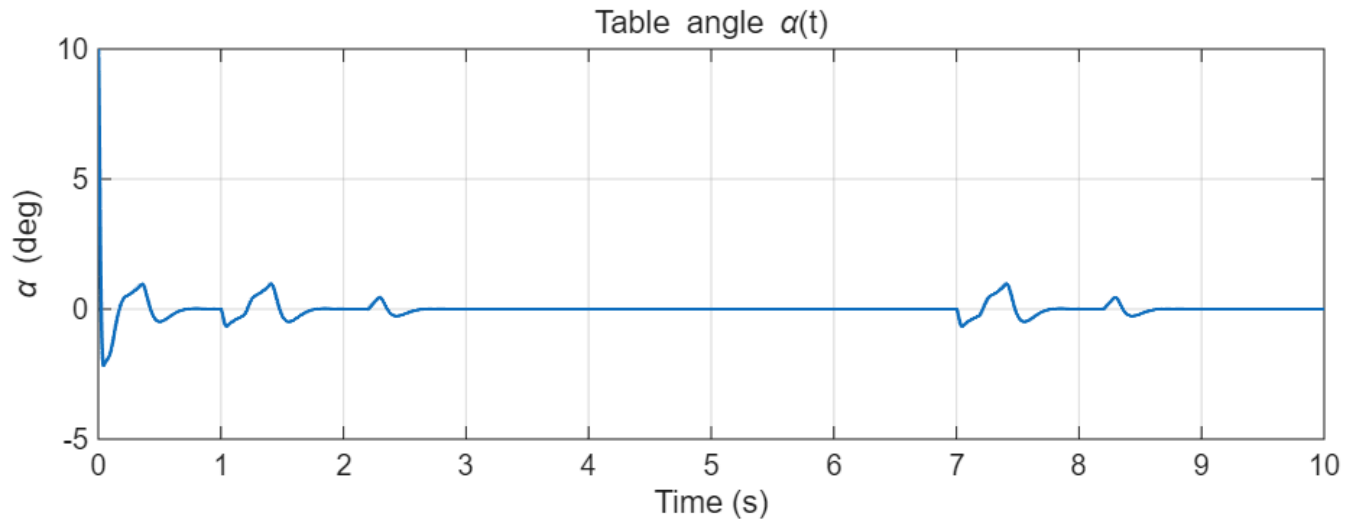
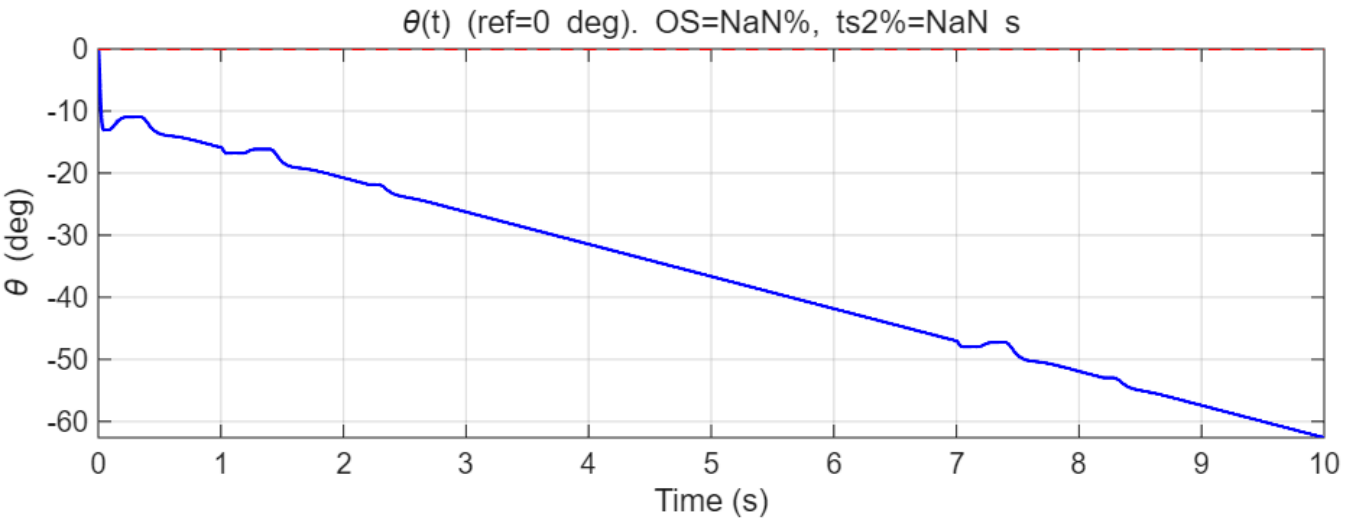
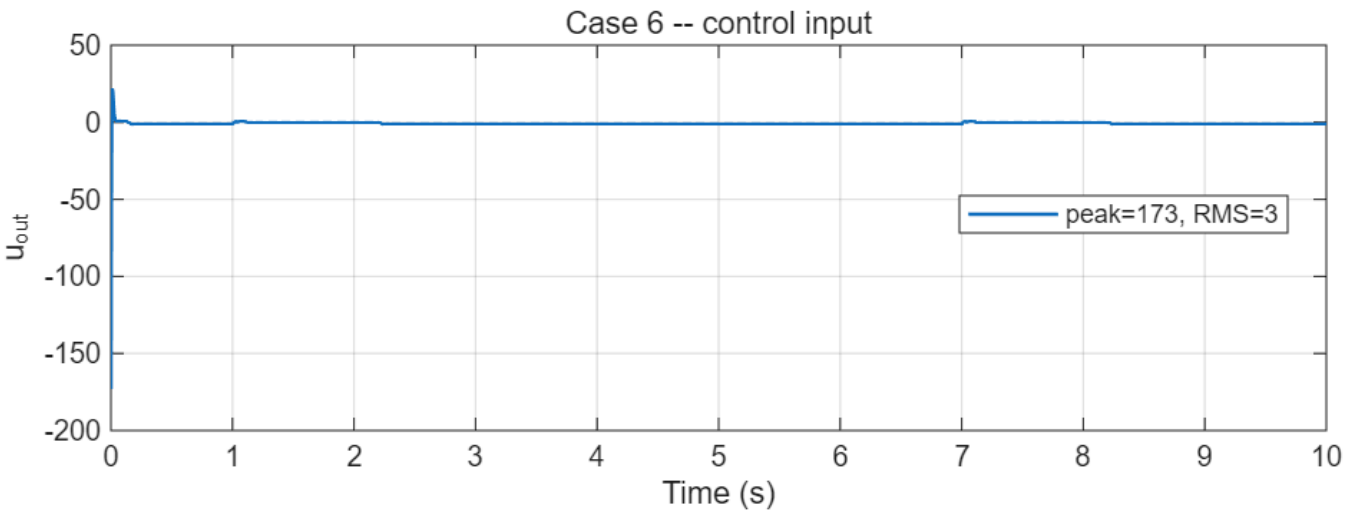


Table angle α(t)
α (deg)
10
5
0
-5
0 1 2 3 4 5 6 7 8 9 10
Time (s)


θ(t) (ref=0 deg). OS=NaN%, ts2%=NaN s
θ (deg)
0
-10
-20
-30
-40
-50
-60
0 1 2 3 4 5 6 7 8 9 10
Time (s)


Case 6 -- control input
u_{out}
peak=173, RMS=3
Time (s)

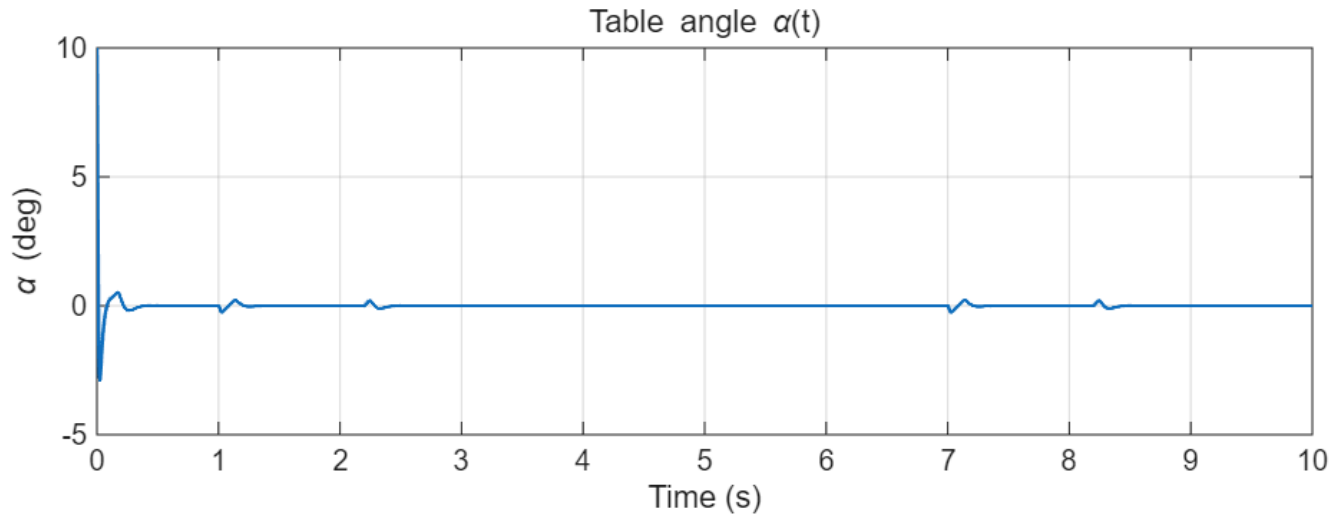
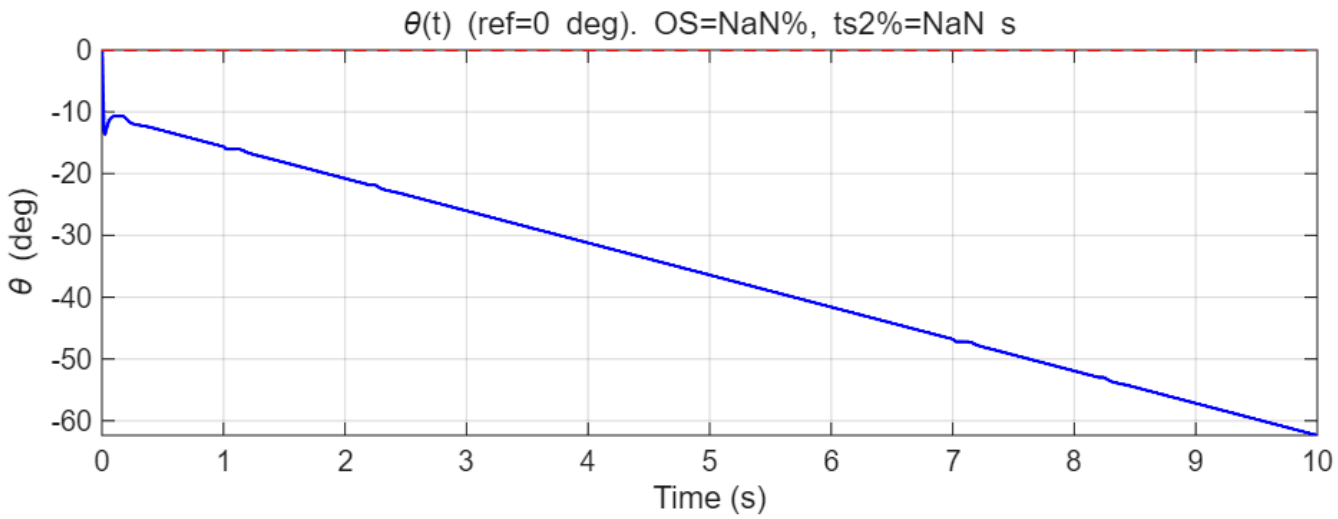
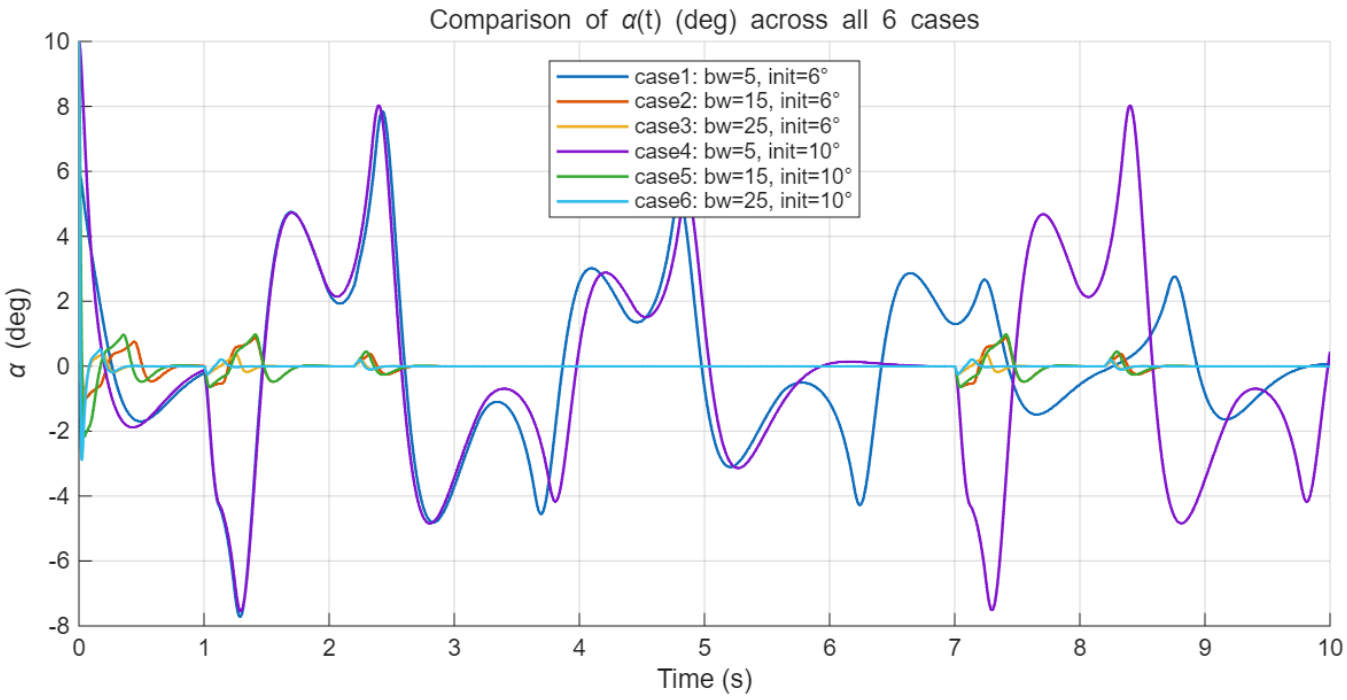
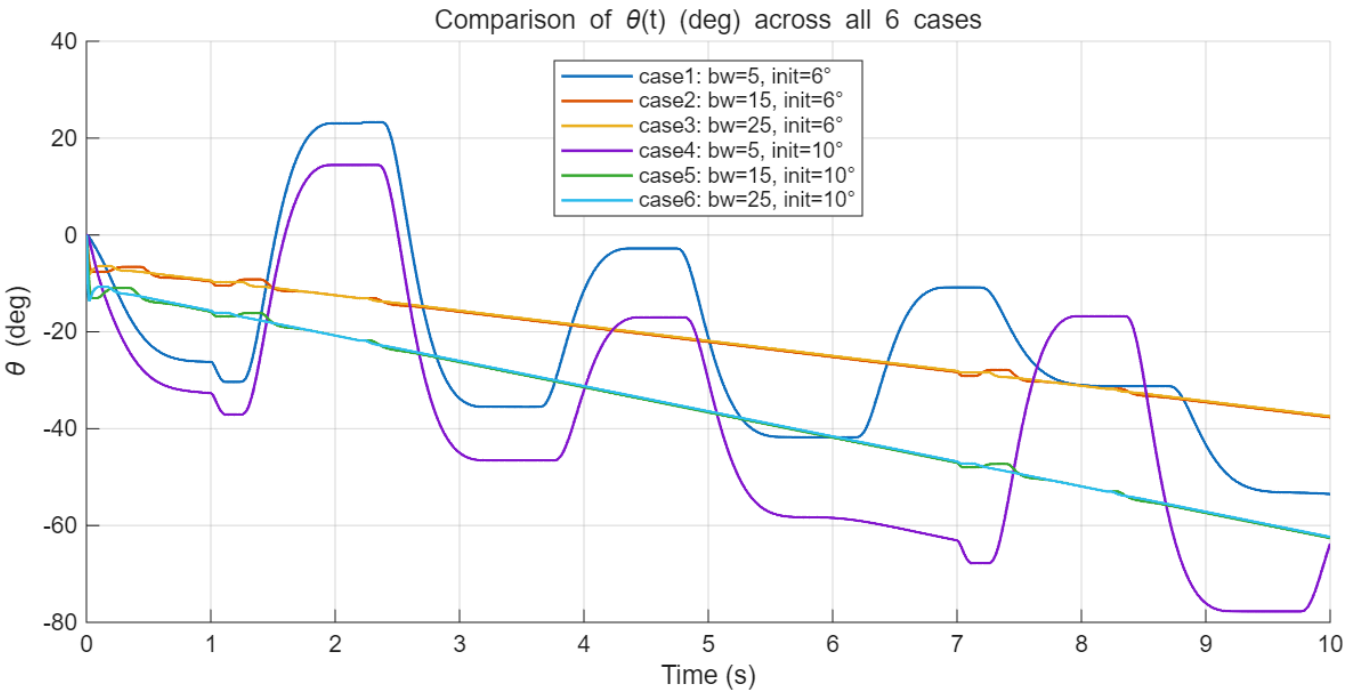


Table angle α(t)
α (deg)
Time (s)


θ(t) (ref=0 deg). OS=NaN%, ts2%=NaN s
θ (deg)
Time (s)

Comparison





Result

The closed-loop bandwidth determines transient speed: larger ω_{cl} produces faster rise and settling times, but also larger control effort and reduced robustness to nonlinearity and actuator limits. In simulations the 25-rad/s designs show the smallest settling times but also the largest control peaks.

When the initial pole angle is larger (10°), larger overshoot and oscillation occur compared to the 6°.